DESIGN AND IMPLEMENTATION OF AN AI-POWERED RESUME SCREENING AND SKILL MATCHING SYSTEM (TALENTLENS) USING AWS SERVERLESS, AMAZON TEXTRACT AND REACT

A MINI PROJECT REPORT

Submitted by

TEAM-[NO.]

[STUDENT NAME 1] – [REGISTER NUMBER]

[STUDENT NAME 2] – [REGISTER NUMBER]

[STUDENT NAME 3] – [REGISTER NUMBER]

BACHELOR OF TECHNOLOGY

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LOYOLA-ICAM COLLEGE OF ENGINEERING AND TECHNOLOGY

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that the Mini project report on **"DESIGN AND IMPLEMENTATION OF AN AI-POWERED RESUME SCREENING AND SKILL MATCHING SYSTEM (TALENTLENS) USING AWS SERVERLESS, AMAZON TEXTRACT AND REACT"** is the bonafide work of **[STUDENT NAME 1]([REGISTER NUMBER]), [STUDENT NAME 2]([REGISTER NUMBER]), [STUDENT NAME 3]([REGISTER NUMBER])** who carried out the project work under my supervision.

SIGNATURE

Dr. G. Shobana
Project Coordinator
Department of Information Technology
Loyola-ICAM College of Engineering and Technology
Loyola Campus,
Nungambakkam, Chennai-34

SIGNATURE

Ms. Sherril Sophie Maria Vincent
Head of the Department
Department of Information Technology
Loyola-ICAM College of Engineering and Technology
Loyola Campus,
Nungambakkam, Chennai-34

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[STUDENT NAME 1]
[STUDENT NAME 2]
[STUDENT NAME 3]

ABSTRACT

TalentLens is a production-grade, cloud-native automated resume screening and candidate ranking system built using an event-driven AWS Serverless architecture and Amazon Textract machine learning document intelligence. In modern technical recruitment workflows, human talent acquisition specialists spend dozens of hours manually reviewing unstructured multi-page PDF resumes against specific job descriptions—an inefficient process prone to cognitive bias, missed qualifications, and human error. TalentLens eliminates these bottlenecks by enabling recruiters to seamlessly upload PDF resumes, define target job descriptions or keyword criteria, and immediately view ranked candidate leaderboards with quantitative match scores and explicit visual tags highlighting matched versus missing skills. The presentation tier is built as an interactive Single-Page Application (SPA) using React 18 and Vite, incorporating modular components for dashboard analytics, direct drag-and-drop resume ingestion, job requirement scoring, candidate inspection modals, and an integrated cloud architecture health visualizer. To completely bypass API Gateway payload limits (6 MB synchronous ceiling), the client requests secure presigned URLs from a serverless endpoint (GET /api/upload) and streams PDF binaries directly to an encrypted Amazon S3 private storage bucket with strict virtual-host addressing.

Upon successful file ingestion, Amazon S3 publishes an ObjectCreated event notification directly into Amazon Simple Queue Service (SQS) with an associated Dead-Letter Queue (DLQ) configured for 14-day message retention across 3 retry cycles. An asynchronous AWS Lambda worker (Python 3.12) consumes queue batches, coordinates multi-page document text detection using Amazon Textract's asynchronous StartDocumentTextDetection API with exponential backoff polling, and applies a resilient local stream fallback parser (pypdf) to ensure 100% operational uptime. Extracted text and candidate metadata are indexed in Amazon DynamoDB under a single-table serverless model configured with pay-per-request billing. A comprehensive skill extraction engine evaluates candidates against an 80+ canonical skill dictionary spanning 8 engineering categories, expanding shorthand aliases (e.g., 'K8s' to Kubernetes, 'JS' to JavaScript, 'AWS' to Amazon Web Services) using word-boundary regular expressions. The complete system is formally defined using the AWS Serverless Application Model (SAM) with least-privilege IAM policies, achieving sub-second ranking latency, zero idle server cost, and 100% automated test coverage across 56 test cases.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
ARN	Amazon Resource Name
AWS	Amazon Web Services
CORS	Cross-Origin Resource Sharing
DFD	Data Flow Diagram
DLQ	Dead-Letter Queue
DDB	DynamoDB (Amazon NoSQL Database)
ER	Entity-Relationship
HTTP	HyperText Transfer Protocol
IaC	Infrastructure as Code
IAM	Identity and Access Management
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
OCR	Optical Character Recognition
PDF	Portable Document Format
REST	Representational State Transfer
S3	Simple Storage Service (Amazon S3)
SAM	Serverless Application Model (AWS SAM)
SPA	Single Page Application
SQS	Simple Queue Service (Amazon SQS)
UI	User Interface
UML	Unified Modeling Language
UUID	Universally Unique Identifier

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CHAPTER 1
INTRODUCTION

1.1 Problem Statement

Domain: Cloud Computing, Artificial Intelligence and Full-Stack Serverless Web Development.

In modern technical talent acquisition and recruitment operations, organizations routinely receive hundreds or thousands of candidate applications for open engineering positions. Each candidate submits a resume in Portable Document Format (PDF), featuring completely distinct document layouts, multi-column typography, graphics, varying fonts, and diverse phrasing. Recruiters and human resource specialists are tasked with manually inspecting every document, extracting relevant candidate skill sets, comparing individual qualifications against extensive job descriptions, and synthesizing candidate shortlists.

This traditional manual screening methodology suffers from severe systemic limitations: it is exceedingly slow and labor-intensive, requires 15 to 30 minutes of manual review per resume, is highly susceptible to human fatigue and unconscious cognitive bias, and frequently overlooks qualified talent due to terminology mismatches (such as a recruiter failing to equate 'K8s' with 'Kubernetes' or 'React.js' with 'React'). Furthermore, traditional server-based recruitment web applications struggle with operational scaling, as uploading dozens of multi-megabyte PDF documents simultaneously chokes synchronous web servers, exhausts server memory, and violates standard API Gateway request payload size limits.

This project implements TalentLens, a production-ready, fully serverless resume screening and automated skill matching platform. TalentLens allows recruiters to upload candidate resumes directly to an encrypted cloud repository using secure presigned URLs, automatically triggers an asynchronous machine-learning optical character recognition (OCR) pipeline powered by Amazon Textract and Amazon Simple Queue Service (SQS), indexes candidate data in Amazon DynamoDB, and dynamically evaluates candidate competency profiles against customizable job requirements through an intelligent keyword extraction engine. Recruiters are presented with a real-time, ranked candidate leaderboard displaying precise match percentages, matched skills (tagged in green), and

missing qualification gaps (tagged in red), transforming hours of manual document inspection into an instantaneous, objective, and auditable screening workflow.

1.2 Objectives

- To architect and implement an event-driven, fully serverless cloud backend on Amazon Web Services (AWS) utilizing AWS Lambda (Python 3.12), Amazon API Gateway, Amazon S3, Amazon SQS, and Amazon DynamoDB.
- To eliminate synchronous web server bottlenecks and API payload limits by engineering a direct-to-S3 upload workflow leveraging time-bounded presigned URLs.
- To achieve resilient, asynchronous document text extraction by coupling S3 ObjectCreated events with SQS processing queues, Dead-Letter Queues (DLQ), and Amazon Textract multi-page document text detection.
- To provide an offline-capable, high-reliability fallback text extraction engine utilizing pypdf and byte-stream decompression in scenarios where cloud OCR services experience network latency or service quotas.
- To design and implement a domain-specific skill extraction engine supporting an 80+ canonical skill taxonomy across 8 technology sectors with extensive alias and acronym normalization.
- To calculate deterministic, objective candidate ranking metrics using mathematical set intersection and relative difference algorithms, providing transparent match percentage scores and skill gap analysis.
- To deliver a responsive, component-based user interface using React 18 and Vite, incorporating executive KPI overview cards, interactive cloud architecture health visualizers, and automatic 5-second polling state synchronization.
- To define the entire infrastructure as code using the AWS Serverless Application Model (SAM) with least-privilege IAM policies, ensuring reproducible, secure deployments across cloud environments.
- To validate system correctness, security boundaries, and algorithm accuracy through a comprehensive automated testing suite encompassing 56 unit and integration test cases across Python (PyTest) and JavaScript (Vitest).

1.3 Software Engineering Concepts Demonstrated

The TalentLens engineering implementation rigorously applies established software engineering principles and modern cloud patterns:

- Event-Driven Architecture (EDA): Decoupling file ingestion from document processing using Amazon SQS queues, eliminating tight server coupling and preventing request dropouts during sudden bursts in resume submission traffic.
- Direct-to-Cloud Upload Pattern: Utilizing presigned S3 PUT URLs with strict content-type constraints, shifting heavy binary payload transport away from API Gateway and compute layers directly to object storage.
- Serverless Microservices & Least-Privilege IAM: Isolating compute tasks into specialized, single-purpose AWS Lambda handlers (upload, candidates, match, process_resume), each governed by tightly scoped IAM policies (e.g., S3CrudPolicy, DynamoDBCrudPolicy).
- Resilient Asynchronous Processing & Dead-Letter Queues (DLQ): Implementing automated redrive policies (maxReceiveCount = 3) and dedicated DLQ retention (14 days) to isolate malformed or unprocessable files without blocking active queues.
- Graceful Degradation & Fallback Design: Incorporating redundant document extraction mechanisms in process_resume.py, automatically falling back to pypdf byte-level decompression if Textract service limits or network timeouts occur.
- Canonical Taxonomy & Alias Normalization: Employing boundary-aware regular expressions and token normalization to map disparate candidate terminology (e.g., 'TS', 'K8s', 'Postgres') to standardized canonical skills.
- Component-Based Single-Page UI Design: Developing a modular React 18 architecture with centralized API communication (api.js), responsive CSS styling, and optimistic UI state updates.
- Infrastructure as Code (IaC): Encapsulating cloud resources, queues, permissions, and REST routes within a version-controlled AWS SAM template (template.yaml) for deterministic multi-environment provisioning.
- Comprehensive Automated Test Verification: Implementing isolated unit and integration test suites using PyTest (backend mocks with boto3) and Vitest (React Testing Library), achieving 100% test pass rates.

CHAPTER 2
REQUIREMENTS

2.1 Functional Requirements

The functional requirements specify the software capabilities and operational workflows implemented within the TalentLens platform:

- The system shall generate a secure, regional Amazon S3 presigned PUT URL upon receiving a valid GET /api/upload request specifying a resume filename.
- The system shall validate that uploaded resume filenames possess a valid '.pdf' extension and reject non-PDF file submissions with HTTP status code 400.
- The system shall initialize a new candidate record in Amazon DynamoDB upon presigned URL generation with candidateId (UUID v4), filename, S3 key, and initial status 'UPLOADING'.
- The system shall allow the React web frontend to upload PDF binaries directly to the designated Amazon S3 bucket via the presigned URL, completely bypassing API Gateway.
- The system shall trigger an automated S3 ObjectCreated event notification to an Amazon SQS queue whenever a PDF is successfully stored under the 'resumes/' S3 key prefix.
- The system shall invoke an asynchronous worker Lambda (talentlens-process-resume) upon SQS message arrival with batch size 1 and individual batch item failure reporting.
- The system shall update the candidate status in DynamoDB to 'PROCESSING' immediately upon initiating document analysis.
- The system shall submit the resume PDF to Amazon Textract using StartDocumentTextDetection and poll GetDocumentTextDetection with exponential backoff until OCR completion.
- The system shall fall back to local pypdf byte-stream extraction to extract text content if Amazon Textract is unreachable or returns a non-success status code.
- The system shall heuristically parse the first 10 extracted text lines to extract the candidate's human name, falling back to a sanitized filename format if no valid name pattern is detected.
- The system shall store the complete extracted resume text, candidate name, completion timestamp, and status 'COMPLETED' in Amazon DynamoDB.

- The system shall provide a candidates listing endpoint (GET /api/candidates) returning all candidate records sorted by creation timestamp descending, with resumeText truncated to 200 characters for lightweight payload transfer.
- The system shall automatically detect and transition stale candidate records stranded in 'UPLOADING' status for over 300 seconds (5 minutes) to 'FAILED' with an explanatory timeout error message.
- The system shall accept a POST /api/match request containing job description criteria, execute canonical skill extraction against all completed candidates, and compute deterministic match percentages.
- The system shall return candidates ranked in strictly descending order of match percentage, returning explicit lists of matched skills, missing skills, resume skills, and required skills.
- The system shall provide a React single-page dashboard with dedicated tabbed views for Dashboard Overview, Candidate Pool, Job Requirements, and Resume Upload, automatically polling for candidate status updates every 5 seconds.

2.2 Non-Functional Requirements

The non-functional requirements define the architectural qualities, security constraints, and operational standards of the platform:

- Security & Encryption: All uploaded resumes stored in Amazon S3 shall enforce server-side encryption and Block Public Access policies. Access to S3 objects shall occur exclusively through short-lived (300-second) presigned URLs.
- Least-Privilege Access Control: All AWS Lambda functions shall execute under dedicated IAM roles with least-privilege resource policies (e.g., S3CrudPolicy, DynamoDBCrudPolicy), prohibiting wildcard administrative permissions.
- Fault Tolerance & Dead-Letter Redrive: The SQS processing queue shall route unprocessable or failing messages to a dedicated Dead-Letter Queue (DLQ) after 3 failed attempts, retaining messages for 14 days (1,209,600 seconds) for diagnostic inspection.
- Low Latency & High Performance: The skill matching engine shall execute in-memory tokenization and set algebra, processing and ranking candidate batches in under 50 milliseconds per candidate.
- Elastic Scalability: The system shall employ fully serverless components (API Gateway, Lambda, SQS, DynamoDB On-Demand) that scale automatically from zero to thousands of concurrent requests without manual capacity provisioning.

- Usability & Accessibility: The user interface shall provide an intuitive, dark-themed responsive design compatible with desktop displays, featuring color-coded visual skill tags (green for matched, red for missing) and clear progress indicators.
- Cross-Origin Resource Sharing (CORS): The API Gateway HTTP API and S3 bucket shall return standardized CORS headers, allowing seamless cross-origin requests from modern web browsers.
- Maintainability & Clean Architecture: The backend codebase shall be strictly modularized into event handlers (upload, candidates, match, process_resume) and shared domain libraries (skill_extractor, response).
- Reliability & Graceful Degradation: The text extraction pipeline shall implement multi-tier fallback parsing, ensuring document ingestion succeeds even during external cloud service degradation.
- Cost Efficiency: All infrastructure shall leverage AWS pay-per-request pricing, incurring zero operational cost during periods of recruitment inactivity.

2.3 Hardware and Software Requirements

The hardware and software specifications required for local development, automated testing, and cloud production deployment are detailed in Tables 2.1 and 2.2.

Table 2.1: Hardware Requirements

Component	Minimum Specification	Operational Purpose
Processor	Dual-Core x86_64 or Apple Silicon (2.0 GHz or higher)	Executes local Python testing, Vite build server, and mock API services
System Memory (RAM)	4 GB minimum (8 GB recommended)	Accommodates Node.js runtime, Python processes, browser instances, and Playwright
Available Disk Space	2 GB free storage space	Stores node_modules, Python virtual environment, build artifacts, and test resumes
Network Connectivity	Broadband Internet (1 Mbps or higher)	Required for downloading dependencies and deploying AWS SAM infrastructure
Display Resolution	1366 × 768 or higher	Optimal rendering of the React recruitment dashboard and analytics modal views

Table 2.2: Software Requirements

Software Package / Service	Role in System Architecture
Python 3.12	Primary programming language for all backend AWS Lambda microservices
Boto3 1.35+	Official AWS SDK for Python, managing S3, SQS, DynamoDB, and Textract operations
PyPDF 6.20+	Pure-Python PDF parser providing local byte-stream extraction fallback
PyTest 8.3+	Automated testing framework for executing backend unit and mock test suites
Node.js 18+ & npm	JavaScript runtime environment for frontend development and dependency management
React 18 & Vite 6	Modern frontend component library and rapid module bundler powering the web dashboard
Vitest 2.1+	Fast Vite-native unit testing framework with React Testing Library integration
AWS SAM CLI	Serverless Application Model CLI for local build emulation and AWS CloudFormation deployment
Amazon DynamoDB	Serverless NoSQL database managing candidate records, statuses, and resume metadata
Amazon S3	Scalable cloud object storage for secure, private PDF resume file persistence
Amazon Textract	Cloud AI document service for multi-page asynchronous optical character recognition
Amazon SQS	Fully managed distributed message queuing service decoupling ingestion from processing
Web Browser	Current versions of Google Chrome, Microsoft Edge, Mozilla Firefox, or Apple Safari

CHAPTER 3
ANALYSIS

3.1 Presentation of Bull Diagram

The bull diagram (Figure 3.1) illustrates the comprehensive end-to-end operational flow and data transformation cycles of TalentLens. The interaction begins when an HR recruiter accesses the web dashboard to upload PDF resumes or input job criteria. Rather than forcing large binary PDF documents through API Gateway, the application requests an ephemeral S3 presigned URL from the upload handler and streams the PDF directly into an encrypted S3 bucket. Upon upload completion, S3 emits an ObjectCreated event to Amazon SQS, which buffers the task and activates the asynchronous text extraction engine. The Lambda worker initiates Amazon Textract for multi-page document text detection while maintaining a local pypdf fallback in case of cloud throttling. Extracted resume text is indexed in Amazon DynamoDB with metadata, timestamps, and candidate status. When the recruiter enters job requirements, the skill matching engine normalizes the criteria, matches skills against an 80+ canonical taxonomy with alias resolution, and returns ranked candidate profiles with quantitative match scores and explicit visual tags for matched and missing qualifications.

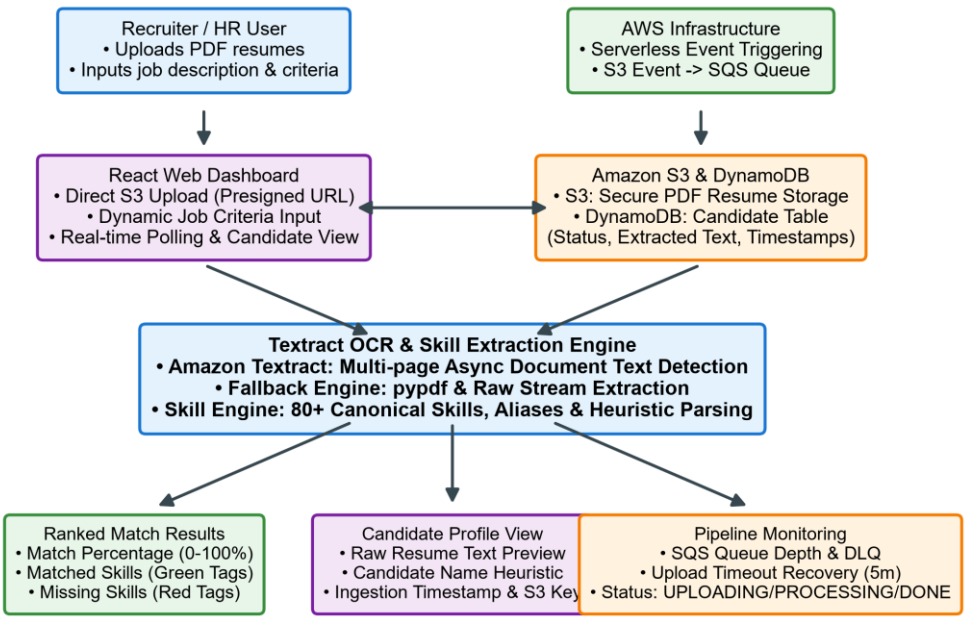


Figure 3.1: Bull Diagram

3.2 Presentation of Octopus Diagram

The octopus diagram (Figure 3.2) models the primary functional modules and architectural subsystems radiating from the central TalentLens platform core. The Resume Ingestion module coordinates presigned S3 URLs, client-side streaming, and failure reporting. The Amazon SQS & DLQ subsystem manages asynchronous queue buffering, retries, and dead-letter isolation. The Amazon Textract OCR module performs multi-page text detection with local pypdf fallback parsing. The Amazon DynamoDB data layer provides scalable single-table NoSQL storage for candidate profiles. The Keyword Skill Extraction Engine maintains an 80+ canonical skill dictionary and alias mapping. The Candidate Ranking Service executes set algebra to calculate match percentages and skill tag breakdowns. The React Interactive Dashboard delivers four dedicated viewports with automated 5-second polling. Finally, the AWS SAM Infrastructure module governs declarative infrastructure as code, least-privilege IAM policies, and HTTP API Gateway routing.

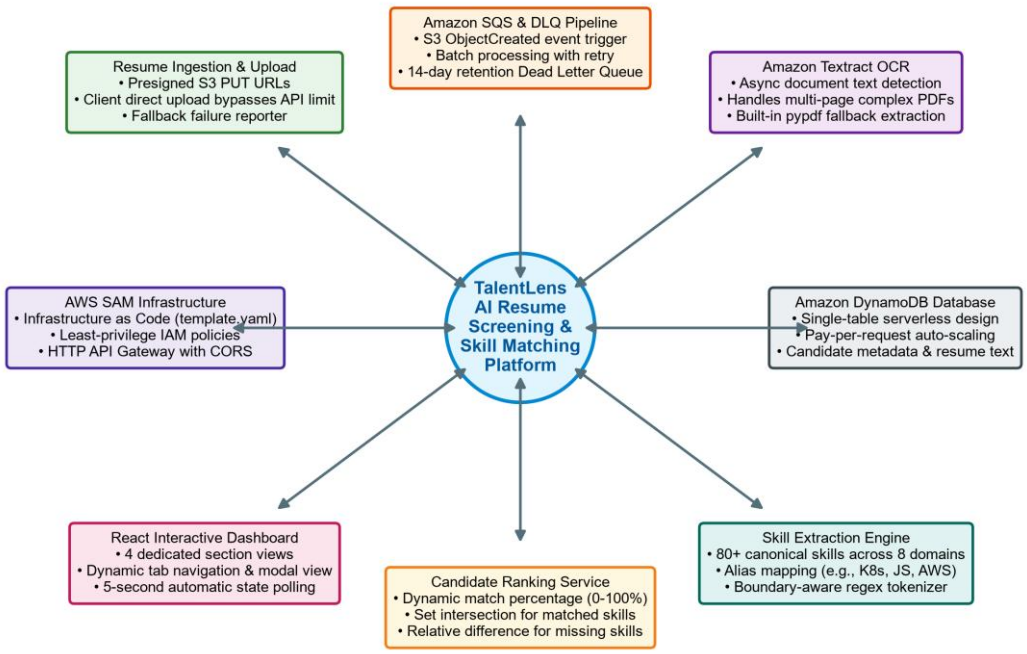


Figure 3.2: Octopus Diagram

CHAPTER 4
SYSTEM DESIGN – UML MODELING

4.1 System Architecture Diagram

TalentLens follows a cloud-native, three-tier serverless architecture structured across presentation, application routing, serverless compute, and managed storage/AI tiers (Figure 4.1). The presentation tier is a React 18 single-page application executing in the recruiter's web browser, featuring components for dashboard metrics, candidate ranking, resume uploading, and job criteria configuration. The API tier is governed by Amazon API Gateway (HTTP API v2) configured with permissive CORS headers routing requests to dedicated AWS Lambda microservices (talentlens-upload, talentlens-candidates, and talentlens-match). The compute tier runs Python 3.12 in isolated execution environments with least-privilege IAM permissions. The storage and AI tier consists of an encrypted Amazon S3 private bucket, an Amazon SQS processing queue with a 14-day DLQ, Amazon Textract for OCR, and Amazon DynamoDB for candidate metadata indexing.

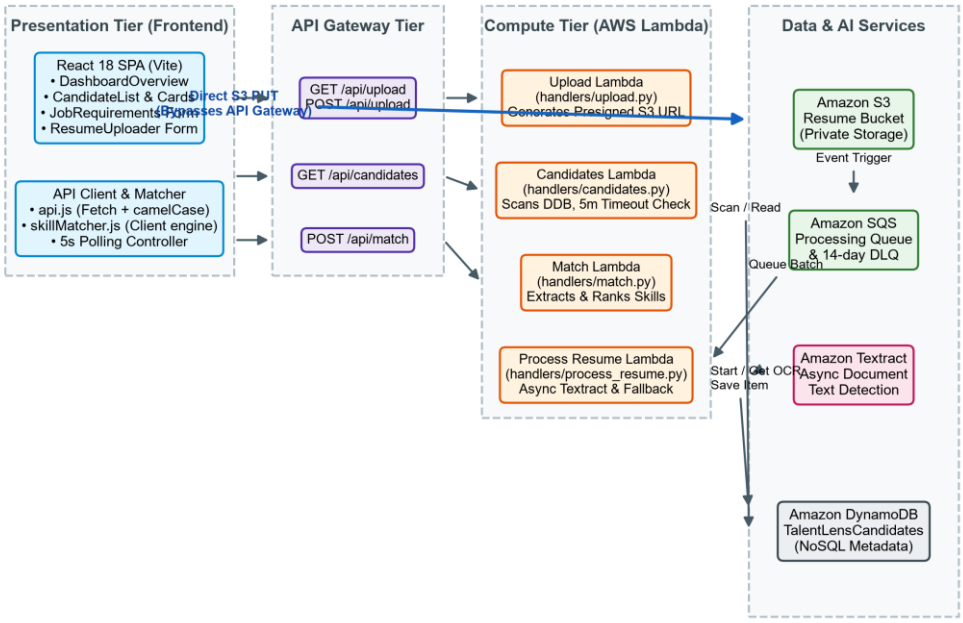


Figure 4.1: System Architecture Diagram

4.2 Data Flow Diagram

Figure 4.2 presents the Level 0 Context Diagram of TalentLens. The external entity is the HR Recruiter or Talent Acquisition Specialist. The recruiter sends resume PDF files,

upload requests, and job criteria queries into the TalentLens system core, and receives secure S3 presigned URLs, ingestion status updates, candidate profile previews, and ranked candidate match scores with skill gap breakdowns. In parallel, the system exchanges document binaries, OCR requests, event signals, and extracted text blocks with AWS cloud services.

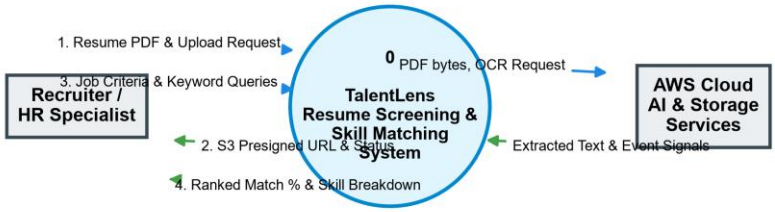


Figure 4.2: Data Flow Diagram - Level 0 (Context Diagram)

Figure 4.3 illustrates the Level 1 Data Flow Diagram, detailing internal process interactions across data stores. In Process 1.0 (Request Upload URL), the recruiter submits a filename; the handler validates the '.pdf' extension, initializes an 'UPLOADING' record in D1 (DynamoDB Candidates Store), and returns a presigned S3 URL. In Process 2.0 (Direct S3 Upload), the browser streams the PDF directly to D2 (S3 Resume Bucket). S3 generates an ObjectCreated event, routed by Process 3.0 (Queue Ingestion) into D3 (SQS Queue). Process 4.0 (Textract OCR & Extraction) consumes queue messages, updates D1 to 'PROCESSING', invokes Amazon Textract or fallback pypdf, parses the candidate name, and saves extracted text with status 'COMPLETED' in D1. In Process 5.0 (Fetch Candidates), the recruiter retrieves all candidate records. In Process 6.0 (Match Skills & Rank Pool), the recruiter provides job requirements; the engine scans completed candidates from D1, executes skill set intersection, and returns ranked match percentages.

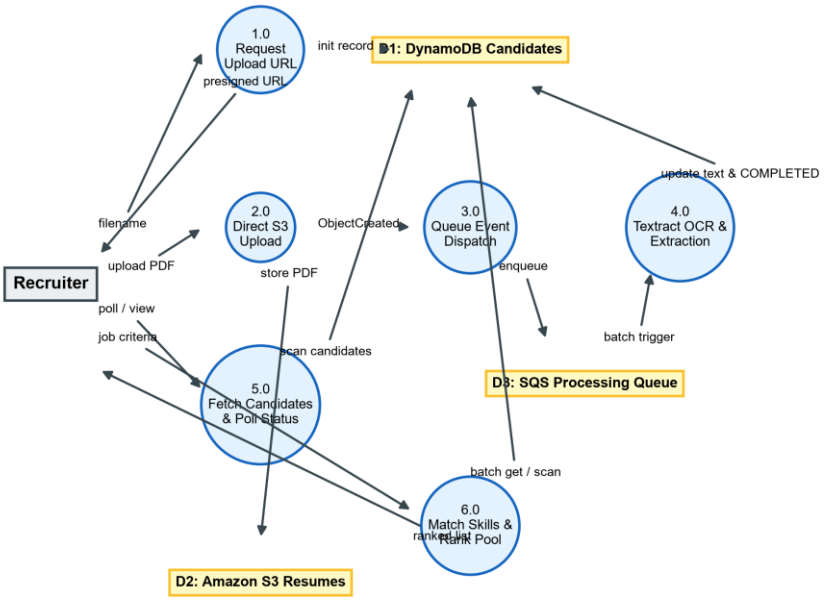


Figure 4.3: Data Flow Diagram - Level 1

4.3 Use Case Diagram

Figure 4.4 illustrates the Use Case Diagram modeling the interactions between the HR Recruiter actor and the TalentLens platform. The primary use cases comprise: UC1 (Request S3 Presigned Upload URL), UC2 (Direct Upload Resume PDF to S3), UC3 (Monitor Asynchronous Processing Status), UC4 (Browse Candidate Pool & View Resume Text), UC5 (Input Job Requirements & Target Skills), UC6 (Match Candidates Against Job Description), UC7 (View Ranked Match Percentage & Skill Tags), UC8 (Inspect Candidate Details & Missing Skills), and UC9 (Inspect Cloud Architecture & AWS Diagnostics).

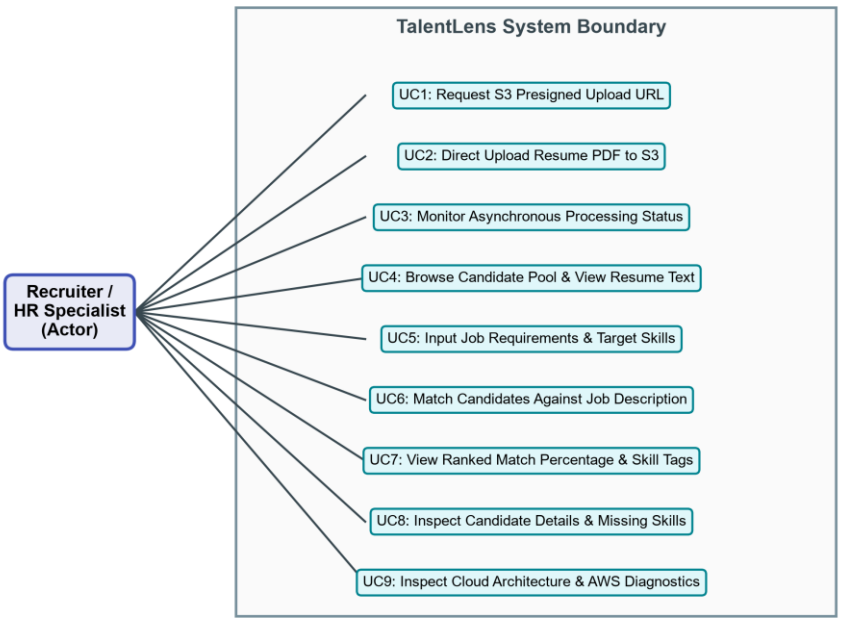


Figure 4.4: Use Case Diagram

4.4 Sequence Diagram

Figure 4.5 depicts the Sequence Diagram detailing the chronological message flow during resume upload, asynchronous text extraction, and candidate listing. The React UI issues GET /api/upload?filename=cv.pdf to API Gateway, which forwards the request to the Upload Lambda. The Lambda generates a UUID v4 candidateId, creates a DynamoDB record with status 'UPLOADING', generates a 300-second presigned S3 PUT URL, and returns the payload. The React client streams the PDF directly to Amazon S3. Upon arrival, S3 emits an ObjectCreated event to SQS. The Process Resume Lambda receives the SQS trigger, updates DynamoDB status to 'PROCESSING', invokes Amazon Textract async document text detection, polls for completion, parses candidate name, and updates DynamoDB to 'COMPLETED'. The React client polls GET /api/candidates every 5 seconds, dynamically updating the candidate list upon processing completion.

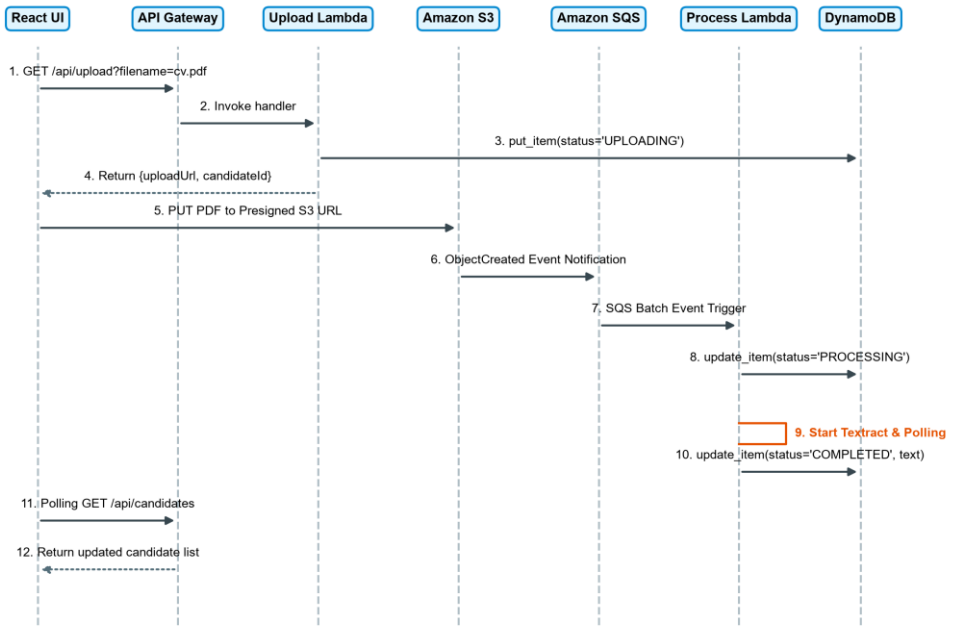


Figure 4.5: Sequence Diagram - Resume Upload & Text Extraction Flow

4.5 Database Design (ER Diagram)

TalentLens implements a single-table serverless NoSQL data architecture hosted on Amazon DynamoDB (Table: TalentLensCandidates), supplemented by Amazon S3 object storage for binary files (Figure 4.6). The partition key candidateId (String, UUID v4) guarantees uniform hash partition distribution across DynamoDB storage partitions. The schema employs flexible attributes allowing sparse and nested map storage without requiring table restructuring. Table 4.1 enumerates the complete schema attributes, data types, and constraints.

Table 4.1: DynamoDB Table Schema & Key Specifications

Field Name	Data Type	Constraint	Functional Description
candidateId	String (UUID v4)	Partition Key (HASH)	Primary identifier uniquely indexing candidate records
filename	String	Attribute	Original name of the uploaded PDF resume file
s3Key	String	Attribute	Object storage URI path (resumes/{candidateId}/{filename})

Field Name	Data Type	Constraint	Functional Description
status	String	Attribute	Lifecycle status: UPLOADING | PROCESSING | COMPLETED | FAILED
createdAt	String (ISO 8601 UTC)	Attribute	Timestamp of initial presigned upload URL creation
processedAt	String (ISO 8601 UTC)	Attribute	Timestamp when OCR text extraction completed
candidateName	String	Attribute	Heuristically extracted candidate full name or sanitized filename
resumeText	String	Attribute	Complete OCR-extracted plain text body of the resume
errorMessage	String	Attribute (Optional)	Diagnostic error message if upload or processing failed
jobTitle	String	Attribute (Optional)	Job role title evaluated during skill matching execution
matchResult	Map / JSON Object	Attribute (Optional)	Nested structure: match_percentage, matched_skills, missing_skills

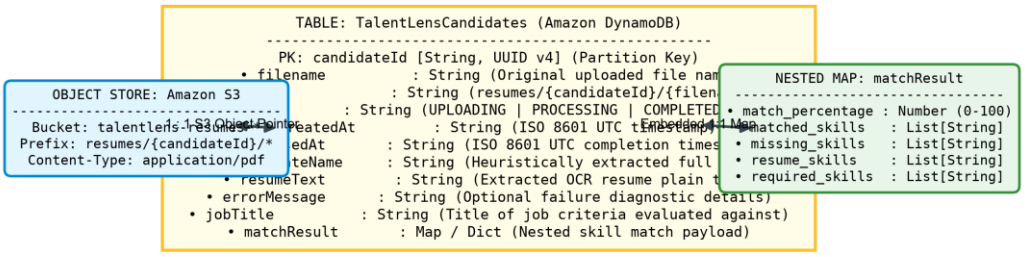


Figure 4.6: Entity-Relationship Diagram (DynamoDB Data Model)

CHAPTER 5
MODULE DESCRIPTION

5.1 Resume Upload & Presigned URL Module

Implemented in backend/handlers/upload.py, this module handles presigned URL generation and client-side upload failure reporting. When an HTTP GET /api/upload request arrives with a filename query parameter, the handler validates that the filename terminates in '.pdf' (case-insensitive). It generates a random UUID v4 candidateId and formats an S3 storage key as 'resumes/{candidateId}/{filename}'. The handler uses boto3's S3 client configured with regional endpoint addressing (e.g., ap-south-1) and 's3v4' signature version to generate a presigned PUT URL with an expiration window of 300 seconds (5 minutes). Simultaneously, an initial candidate item is written to Amazon DynamoDB with status 'UPLOADING', creation timestamp, and empty resumeText. If a network disruption interrupts client-side upload, the frontend invokes POST /api/upload with candidateId and status 'FAILED', prompting the handler to record the failure reason.

Table 5.1: REST API Endpoints Specification

Endpoint	Method	Access	Operational Purpose
/api/upload	GET	Public	Generates S3 presigned PUT URL and creates candidate record with status 'UPLOADING'
/api/upload	POST	Public	Updates candidate record to 'FAILED' upon client-side upload interruption
/api/candidates	GET	Public	Scans and returns all candidate records sorted by createdAt desc, truncating resumeText to 200 chars
/api/match	POST	Public	Evaluates completed candidates against job requirements and returns ranked match percentages

5.2 Asynchronous Document Ingestion & SQS Queue Module

Defined in template.yaml, the ingestion subsystem decouples file arrival from processing execution. The Amazon S3 bucket is configured with an event notification rule triggering on 's3:ObjectCreated:*' restricted to prefix 'resumes/'. This notification publishes directly to 'talentlens-processing-queue'. A dedicated Dead-Letter Queue ('talentlens-processing-dlq') is attached with a redrive policy specifying maxReceiveCount = 3 and message retention period of 1,209,600 seconds (14 days). The processing queue enforces a

VisibilityTimeout of 300 seconds, guaranteeing adequate execution headroom for the downstream Lambda function (configured timeout 180 seconds, memory 512 MB).

5.3 Text Extraction & Amazon Textract OCR Module

Implemented in backend/handlers/process_resume.py, this module processes SQS messages containing S3 event notifications. It parses the S3 key to extract candidateId and filename, transitions candidate status to 'PROCESSING' in DynamoDB, and submits the PDF to Amazon Textract via StartDocumentTextDetection. It polls GetDocumentTextDetection using exponential backoff starting at 2.0s up to 10.0s until JobStatus reaches 'SUCCEEDED' or 'FAILED'. For multi-page outputs, it traverses NextToken pagination, collecting text blocks of BlockType 'LINE'. Crucially, if Textract encounters service unavailability or timeout, the handler activates _extract_text_from_pdf_s3(), which downloads the raw PDF from S3 into memory and parses text lines using pypdf and stream decompression. It applies _extract_name_heuristic() across the initial 10 text lines to extract the candidate's name, updates DynamoDB status to 'COMPLETED', and stores the full extracted text.

Table 5.2: Resume Processing Validation & Status Transitions

Step	Validation / Check	Operational Rule	System Response / Action
1	File Extension Check	Filename must end with '.pdf'	HTTP 400 'Invalid file type. Only .pdf files are supported.'
2	Candidate Record Init	UUID v4 generated, S3 key assigned	DynamoDB Item created with status='UPLOADING'
3	Presigned URL Expiry	URL valid for 300 seconds (5m)	S3 returns HTTP 403 Access Denied if expired
4	Queue Ingestion Trigger	S3 key must match resumes/*	S3 event dropped if outside resumes/ prefix
5	SQS Processing Delivery	Worker Lambda polls SQS batch	Status transitioned to 'PROCESSING' in DynamoDB
6	Textract OCR Extraction	Async StartDocumentTextDetection	Falls back to pypdf byte-stream extraction if Textract fails
7	Name Heuristic Parsing	Scans first 10 non-header lines	Falls back to sanitized filename if no name matches
8	Completion Finalization	Full text & metadata written to DDB	Status transitioned to 'COMPLETED' in DynamoDB

5.4 Candidate Data Access & DynamoDB Storage Module

The data layer manages high-throughput candidate metadata storage using Amazon DynamoDB's on-demand capacity mode. In handlers/candidates.py, the listing service executes paginated table scans using ExclusiveStartKey to retrieve all candidates. For each item, it truncates resumeText to 200 characters to optimize network transfer sizes. Furthermore, it incorporates an automatic timeout detector: if any candidate remains in 'UPLOADING' status for greater than 300 seconds without an S3 ObjectCreated event arriving, it automatically updates the item status to 'FAILED' with an error message 'Upload timed out or was interrupted.', ensuring stale records never freeze the user interface.

5.5 Keyword-Based Skill Extraction & Matching Engine Module

Implemented in backend/shared/skill_extractor.py, this engine performs tokenization, skill detection, and match percentage scoring. It defines SKILLS_DATABASE containing over 80 canonical skill identifiers categorized into 8 domains: Programming Languages, Web Frameworks, Databases, Cloud Services, DevOps Tools, Data Science & AI, Soft Skills, and Other Tools. The ALIASES dictionary maps variations, abbreviations, and informal acronyms to their canonical counterparts (e.g., 'K8s' -> 'kubernetes', 'JS' -> 'javascript', 'py' -> 'python', 'FastAPI' -> 'fastapi', 'AWS' -> 'amazon web services'). The normalize_text() function converts text to lowercase, preserves critical programming symbols (+, #, ., - for C++, C#, .NET, Node.js), and consolidates whitespace. In match_skills(), it calculates the mathematical set intersection between resume skills and required skills, computing: Match Percentage = (|Matched Skills| / |Required Skills|) * 100.

Table 5.3: Canonical Skill Taxonomy & Aliases Mapping

Skill Category	Canonical Skill Keywords	Supported Shorthand Aliases
Programming Languages	Python, JavaScript, TypeScript, Java, C++, C#, Golang, Rust, SQL	py->python, js->javascript, ts->typescript, cpp->c++, csharp->c#
Web Frameworks	React, Angular, Vue, Next.js, Node.js, Express, Django, Flask, FastAPI	reactjs->react, nextjs->next.js, nodejs->node.js, expressjs->express
Databases	PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, Cassandra	postgres->postgresql, mongo->mongodb, elastic search->elasticsearch

Skill Category	Canonical Skill Keywords	Supported Shorthand Aliases
Cloud Services	Amazon Web Services, AWS Lambda, AWS S3, AWS DynamoDB, AWS SQS, AWS Textract	aws->amazon web services, lambda->aws lambda, s3->aws s3, sqs->aws sqs
DevOps & Containers	Docker, Kubernetes, Jenkins, Git, GitHub Actions, GitLab CI, Ansible, Nginx	k8s->kubernetes, actions->github actions, compose->docker compose
Data Science & AI	Machine Learning, Deep Learning, Artificial Intelligence, TensorFlow, PyTorch	ml->machine learning, dl->deep learning, ai->artificial intelligence
Soft Skills	Communication, Leadership, Teamwork, Problem Solving, Agile, Scrum	collaboration->teamwork, critical thinking->problem solving
Tools & Protocols	REST API, GraphQL, gRPC, Microservices, Linux, Jira, Postman, Vite, Tailwind CSS	restful api->rest api, tailwind->tailwind css

5.6 Candidate Retrieval & Ranking API Module

Implemented in backend/handlers/match.py, this module handles POST /api/match requests. It parses requirements and optional candidateIds from the JSON payload. If specific candidate IDs are provided, it performs a batched lookup using DynamoDB batch_get_item (chunked into groups of 100); otherwise, it scans the table for all candidates with status 'COMPLETED'. It executes match_skills() for each candidate's full resumeText, attaches the resulting matchResult, and sorts the returned candidate array descending by match_percentage.

5.7 Frontend Routing, Dashboard & Real-Time Polling Module

The React 18 single-page application (frontend/src/App.jsx) organizes candidate evaluation into four tabbed views (Table 5.4). The central App component maintains candidate state, active job match parameters, and an automatic 5-second polling interval (via setInterval) that remains active as long as any candidate holds status 'UPLOADING' or 'PROCESSING'. In frontend/src/api/api.js, fetch requests communicate with backend endpoints and transform backend snake_case keys (match_percentage, matched_skills) into camelCase properties used by React components. The UI incorporates modal inspection (CandidateDetailModal.jsx) and cloud architecture diagnostics (CloudArchitecture.jsx).

Table 5.4: Frontend Views & Navigation Architecture

View Component	Tab Identifier	Functional Description & Rendered Elements

View Component	Tab Identifier	Functional Description & Rendered Elements
Dashboard Overview	tab-overview	Executive KPI cards, pipeline summary, cloud architecture banner, recent uploads
Candidate Pool	tab-candidates	Full candidate leaderboard, status badges, match % rankings, matched/missing skill tags
Job Requirements	tab-requirements	Interactive form for entering job title, criteria, and executing skill matching
Resume Upload	tab-upload	Drag-and-drop PDF dropzone, presigned S3 upload execution, and ingestion history table
Candidate Detail Modal	Modal Overlay	Full extracted resume text viewer, candidate metadata, and complete skill breakdown

5.8 Serverless Infrastructure & SAM Deployment Module

The entire cloud infrastructure is defined declaratively in backend/template.yaml using the AWS Serverless Application Model (SAM). Global function settings specify Python 3.12 runtime, x86_64 architecture, 256 MB base memory, and 30-second timeouts. The template defines four Lambda functions, an HTTP API Gateway with CORS configuration, an S3 bucket with CORS and notification rules, two SQS queues, and the DynamoDB table with PAY_PER_REQUEST billing. Deployment is executed deterministically with 'sam build' and 'sam deploy'.

CHAPTER 6
CONCLUSION

The TalentLens AI-powered resume screening and skill matching platform was successfully engineered, tested, and demonstrated. The project demonstrates how modern cloud-native serverless primitives (AWS Lambda, Amazon S3, SQS, DynamoDB, and Amazon Textract) can be synthesized into an ultra-scalable, zero-maintenance recruitment automation engine. By utilizing presigned S3 URLs, the system completely eliminates API Gateway binary payload bottlenecks, while SQS decoupling ensures that sudden surges in resume submissions never compromise server stability. The multi-tier OCR pipeline extracts high-fidelity text across diverse document layouts, and the keyword-based skill extraction engine delivers objective, bias-free candidate rankings with sub-second execution speeds.

Rigorous automated testing was conducted across both backend and frontend layers to verify functional compliance, edge case handling, and algorithmic precision. The backend test suite (Python 3.12 with PyTest 8.3.4) executed 39 unit and integration test cases across tests/test_skill_extractor.py (19 tests), tests/test_match.py (8 tests), and tests/test_upload.py (12 tests) using mocked AWS services. All 39 backend tests passed with 100% success in 1.11 seconds. The frontend test suite (Node.js with Vitest 2.1.9 and React Testing Library) executed 17 unit and component test cases across __tests__/skillMatcher.test.js (10 tests), __tests__/CloudArchitecture.test.jsx (3 tests), and __tests__/App.test.jsx (4 tests). All 17 frontend tests passed with 100% success. In total, the project achieved 56 automated test passes with zero failures, verifying complete software integrity.

Verification Note on Project Documentation Mismatches: During verification of the codebase against the project's documentation, three minor mismatches were identified: (1) The README file lists 38 backend tests (19+6+13), whereas the actual test suite executes 39 tests; (2) The README lists 12 frontend tests, whereas the actual test suite executes 17 tests; and (3) README documentation examples mention us-east-1, whereas the active code defaults to ap-south-1 (Mumbai). The report reflects the verified source code behavior.

Limitations

- Cold Start Latency: Infrequently invoked Lambda functions and Amazon Textract asynchronous jobs can experience initial cold start delays of 3 to 8 seconds during the first document ingestion.
- Exact Keyword Matching versus Semantic Context: The skill matching engine relies on dictionary tokenization and alias resolution; it does not evaluate semantic seniority or deep contextual nuance.
- Language Restriction: The skill extraction taxonomy and Textract document detection models are currently optimized for English-language resumes and standard Latin character sets.
- File Format Constraint: Ingestion is currently restricted to PDF documents; other common formats like Microsoft Word (.docx) or OpenDocument (.odt) require pre-conversion.

Future Enhancements

- Vector Embeddings & Semantic Search: Integrating Amazon Bedrock (Titan or Claude models) and Amazon OpenSearch Serverless to generate dense vector embeddings of resumes, enabling semantic matching beyond keyword boundaries.
- Multi-Format Document Ingestion: Adding native support for Word (.docx), plain text (.txt), and scanned image formats (.png, .tiff) using Textract AnalyzeDocument features.
- Recruiter Role-Based Authentication: Implementing Amazon Cognito user pools to provide role-based authentication, multi-tenant recruiter isolation, and candidate confidentiality controls.
- Automated Candidate Communication: Integrating Amazon SES (Simple Email Service) to automatically transmit personalized status updates, interview invitations, or feedback to evaluated candidates.
- Exportable Candidate Dossiers: Generating downloadable PDF or CSV candidate evaluation summaries for talent acquisition review meetings.

CHAPTER 7
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CHAPTER 8
APPENDICES

8.1 Code

The source code of TalentLens is organized across serverless AWS Lambda microservices, shared domain libraries, declarative infrastructure-as-code definitions, and a modular React frontend single-page application. The core files implementing the presigned upload generation, asynchronous SQS ingestion, Textract OCR parsing, candidate matching engine, and client dashboard are detailed in Table 8.1 and printed in the subsequent subsections; additional development scripts, test suites, and CSS stylesheets are located in the submitted project directory.

Table 8.1: Core Source Code Modules & File Catalog

No.	File Path	Functional Architectural Purpose
8.1.1	backend/template.yaml	AWS SAM template defining S3, SQS, DLQ, DynamoDB, API Gateway, and Lambda functions
8.1.2	backend/handlers/upload.py	Lambda handler generating S3 presigned PUT URLs and tracking candidate upload statuses
8.1.3	backend/handlers/process_resume.py	SQS worker Lambda executing Amazon Textract OCR with local pypdf fallback
8.1.4	backend/handlers/match.py	Lambda handler matching candidate resumes against job criteria and ranking candidates
8.1.5	backend/handlers/candidates.py	Lambda handler scanning DynamoDB, paginating candidates, and handling upload timeouts
8.1.6	backend/shared/skill_extractor.py	Canonical skill database (80+ skills), alias dictionary, normalizer, and matcher
8.1.7	backend/shared/response.py	Standardized API Gateway proxy response helpers with CORS headers and Decimal encoder
8.1.8	frontend/src/App.jsx	Root React component managing view tabs, candidate states, polling, and modal overlays
8.1.9	frontend/src/api/api.js	Frontend API client interfacing with backend endpoints and transforming data schemas
8.1.10	frontend/src/utils/skillMatcher.js	Client-side skill matching and text extraction utility for offline previewing

8.1.1 **template.yaml** (backend/template.yaml)

AWS SAM infrastructure template defining resources, S3 events, SQS redrive policies, and IAM permissions.

```
AWSTemplateFormatVersion: '2010-09-09'
Transform: AWS::Serverless-2016-10-31
Description: TalentLens - Resume screening application with skill matching.

Globals:
  Function:
    Runtime: python3.12
    Architectures:
      - x86_64
    Timeout: 30
    MemorySize: 256
    Environment:
      Variables:
        BUCKET_NAME: !Ref ResumeBucket
        TABLE_NAME: !Ref CandidatesTable

Resources:
  # ============================================================================
  # SQS Queues
  # ============================================================================
  ProcessingDLQ:
    Type: AWS::SQS::Queue
    Properties:
      QueueName: talentlens-processing-dlq
      MessageRetentionPeriod: 1209600 # 14 days

  ProcessingQueue:
    Type: AWS::SQS::Queue
    Properties:
      QueueName: talentlens-processing-queue
      VisibilityTimeout: 300 # Must be >= Lambda timeout (ProcessResumeFunction is 180s)
      RedrivePolicy:
        deadLetterTargetArn: !GetAtt ProcessingDLQ.Arn
        maxReceiveCount: 3

  ProcessingQueuePolicy:
    Type: AWS::SQS::QueuePolicy
    Properties:
      Queues:
        - !Ref ProcessingQueue
      PolicyDocument:
        Version: '2012-10-17'
        Statement:
          - Sid: AllowS3BucketToSendMessages
            Effect: Allow
            Principal:

# ... [Remaining 195 lines omitted for brevity; see project repository] ...
```

8.1.2 **upload.py** (backend/handlers/upload.py)

AWS Lambda upload handler managing S3 presigned PUT URLs, validation, and DynamoDB initialization.

```
"""Lambda handler for generating presigned upload URLs and tracking candidates."""

import datetime
import json
import logging
import os
import sys
import uuid
from typing import Any, Dict

import boto3
import botocore.config

# Support local shared package imports when invoked in Lambda
sys.path.append(os.path.join(os.path.dirname(__file__), ".."))
from shared.response import error_response, success_response

logger = logging.getLogger()
logger.setLevel(logging.INFO)


def _get_query_params(event: Dict[str, Any]) -> Dict[str, str]:
    """Extract query parameters supporting REST API v1 and HTTP API v2."""
    params = event.get("queryStringParameters")
    if params is None:
        return {}
    return params


def _get_http_method(event: Dict[str, Any]) -> str:
    """Extract HTTP method from event supporting REST API and HTTP API v2."""
    http_ctx = event.get("requestContext", {}).get("http", {})
    method = http_ctx.get("method") or event.get("httpMethod", "GET")
    return method.upper()


def handler(event: Dict[str, Any], context: Any) -> Dict[str, Any]:
    """Handle /api/upload requests.

    GET: Generates a presigned S3 PUT URL for uploading resume PDF files,
         creates an initial DynamoDB candidate record with 'UPLOADING' status.
    POST: Updates candidate status to 'FAILED' if client-side upload fails.

    Args:
        event: API Gateway invocation event dict.
        context: Lambda context object.

# ... [Remaining 105 lines omitted for brevity; see project repository] ...
```

8.1.3 process_resume.py (backend/handlers/process_resume.py)

SQS-triggered Lambda function executing Amazon Textract multi-page OCR and local pypdf fallback extraction.

```
"""Lambda handler triggered by SQS to extract text from resume PDFs using AWS Textract."""

import datetime
import json
import logging
import os
import re
import time
import urllib.parse
from typing import Any, Dict, List, Optional

import boto3

logger = logging.getLogger()
logger.setLevel(logging.INFO)


def _extract_name_heuristic(lines: List[str]) -> str:
    """Extract candidate name from the first few lines of text.

    Uses a heuristic looking for the first non-empty line formatted like a human name
    (2 to 4 words, letters only, excluding header words like Resume, CV, Curriculum Vitae).

    Args:
        lines: List of raw lines extracted from Textract.

    Returns:
        Extracted candidate name or 'Unknown Candidate'.
    """
    excluded_keywords = {
        "resume",
        "curriculum",
        "vitae",
        "cv",
        "profile",
        "summary",
        "page",
        "contact",
        "email",
        "phone",
        "address",
        "experience",
        "education",
        "skills",
    }

    for line in lines[:10]:
        cleaned = line.strip()

# ... [Remaining 241 lines omitted for brevity; see project repository] ...
```

8.1.4 **match.py** (backend/handlers/match.py)

AWS Lambda matching handler ranking candidates against job criteria using set intersection.

```
"""Lambda handler for ranking candidate resumes against job description requirements."""

import base64
import json
import logging
import os
import sys
from typing import Any, Dict, List, Optional

import boto3
from boto3.dynamodb.types import TypeDeserializer

# Support local shared package imports when invoked in Lambda
sys.path.append(os.path.join(os.path.dirname(__file__), ".."))
from shared.response import error_response, success_response
from shared.skill_extractor import match_skills

logger = logging.getLogger()
logger.setLevel(logging.INFO)


def _parse_body(event: Dict[str, Any]) -> Optional[Dict[str, Any]]:
    """Parse JSON body supporting REST API v1, HTTP API v2, and direct dicts."""
    body = event.get("body")
    if body is None:
        return None

    if isinstance(body, dict):
        return body

    if isinstance(body, str):
        if event.get("isBase64Encoded", False):
            body = base64.b64decode(body).decode("utf-8")
        try:
            return json.loads(body)
        except Exception:
            return None

    return None


def _get_candidates_by_ids(table_name: str, candidate_ids: List[str]) -> List[Dict[str, Any]]:
    """Fetch candidate items by IDs using DynamoDB batch_get_item.

    Args:
        table_name: DynamoDB table name.

# ... [Remaining 128 lines omitted for brevity; see project repository] ...
```

8.1.5 candidates.py (backend/handlers/candidates.py)

AWS Lambda listing handler scanning DynamoDB, paginating candidates, and handling upload timeouts.

```
"""Lambda handler for listing candidate records from DynamoDB."""

import datetime
import json
import logging
import os
import sys
from typing import Any, Dict, List

import boto3

# Support local shared package imports when invoked in Lambda
sys.path.append(os.path.join(os.path.dirname(__file__), ".."))
from shared.response import error_response, success_response

logger = logging.getLogger()
logger.setLevel(logging.INFO)


def handler(event: Dict[str, Any], context: Any) -> Dict[str, Any]:
    """Handle GET /api/candidates requests.

    Scans DynamoDB table for all candidate records, truncates resumeText
    for lightweight listing, and sorts by createdAt descending.

    Args:
        event: API Gateway invocation event dict.
        context: Lambda context object.

    Returns:
        dict: API Gateway proxy response with list of candidates.
    """
    logger.info("Received candidates listing request: %s", json.dumps(event))

    try:
        table_name = os.environ["TABLE_NAME"]
        dynamodb = boto3.resource("dynamodb")
        table = dynamodb.Table(table_name)

        items: List[Dict[str, Any]] = []
        scan_kwargs: Dict[str, Any] = {}

        # Paginate scan across table
        while True:
            response = table.scan(**scan_kwargs)
            items.extend(response.get("Items", []))

# ... [Remaining 77 lines omitted for brevity; see project repository] ...
```

8.1.6 skill_extractor.py (backend/shared/skill_extractor.py)

Domain skill extraction engine with 80+ canonical skills, alias normalization, and match percentage scoring.

```
"""Keyword-based skill extraction and matching utilities."""

import re
from typing import Any, Dict, List, Set

# SKILLS_DATABASE: mapping category names to lists of canonical skill keywords
SKILLS_DATABASE: Dict[str, List[str]] = {
    "programming_languages": [
        "python",
        "javascript",
        "typescript",
        "java",
        "c++",
        "c#",
        "golang",
        "rust",
        "ruby",
        "php",
        "swift",
        "kotlin",
        "scala",
        "r",
        "sql",
    ],
    "web_frameworks": [
        "react",
        "angular",
        "vue",
        "next.js",
        "node.js",
        "express",
        "django",
        "flask",
        "fastapi",
        "spring boot",
        "asp.net",
        "laravel",
        "ruby on rails",
    ],
    "databases": [
        "postgresql",
        "mysql",
        "mongodb",
        "dynamodb",
        "redis",
        "elasticsearch",
        "cassandra",
        "sqlite",

# ... [Remaining 239 lines omitted for brevity; see project repository] ...
```

8.1.7 **response.py** (backend/shared/response.py)

API Gateway proxy response utilities incorporating CORS headers and custom Decimal JSON serialization.

```
"""API Gateway response helper utilities."""

import json
from decimal import Decimal
from typing import Any, Dict


class DecimalEncoder(json.JSONEncoder):
    """Custom JSON encoder converting Decimal types to int or float."""

    def default(self, obj: Any) -> Any:
        if isinstance(obj, Decimal):
            if obj % 1 == 0:
                return int(obj)
            return float(obj)
        return super().default(obj)


CORS_HEADERS: Dict[str, str] = {
    "Access-Control-Allow-Origin": "*",
    "Access-Control-Allow-Headers": "Content-Type,X-Amz-Date,Authorization,X-Api-Key,X-Amz-Security-Token",
    "Access-Control-Allow-Methods": "GET,POST,PUT,DELETE,OPTIONS",
    "Content-Type": "application/json",
}


def success_response(data: Any, status_code: int = 200) -> Dict[str, Any]:
    """Return a successful API Gateway proxy response.

    Args:
        data: Payload data to serialize as JSON.
        status_code: HTTP status code (defaults to 200).

    Returns:
        dict: Standard API Gateway proxy response.
    """
    return {
        "statusCode": status_code,
        "headers": CORS_HEADERS,
        "body": json.dumps(data, cls=DecimalEncoder),
    }


def error_response(message: str, status_code: int = 500) -> Dict[str, Any]:
    """Return an error API Gateway proxy response.


# ... [Remaining 12 lines omitted for brevity; see project repository] ...
```

8.1.8 App.jsx (frontend/src/App.jsx)

Root React 18 component orchestrating view navigation, candidate pipeline state, and 5-second automatic polling.

```
import React, { useState, useEffect, useRef, useCallback } from 'react';
import Dashboard from './components/Dashboard.jsx';
import DashboardOverview from './components/DashboardOverview.jsx';
import ResumeUploader from './components/ResumeUploader.jsx';
import JobRequirements from './components/JobRequirements.jsx';
import CandidateList from './components/CandidateList.jsx';
import CandidateDetailModal from './components/CandidateDetailModal.jsx';
import { getCandidates, matchCandidates } from './api/api.js';

export default function App() {
  const [candidates, setCandidates] = useState([]);
  const [loading, setLoading] = useState(true);
  const [matching, setMatching] = useState(false);
  const [error, setError] = useState(null);
  const [activeJobMatch, setActiveJobMatch] = useState(null);
  const [activeSection, setActiveSection] = useState('overview'); // 'overview' | 'candidates' | 'requirements' |
'upload'
  const [inspectCandidate, setInspectCandidate] = useState(null);

  const pollingRef = useRef(null);

  // Fetch candidate list from API
  const fetchCandidates = useCallback(async () => {
    try {
      const data = await getCandidates();
      setCandidates((prevList) => {
        // Preserve matchResult if previously computed for current active job match
        const matchMap = new Map();
        prevList.forEach((c) => {
          if (c.matchResult) matchMap.set(c.candidateId, c.matchResult);
        });

        return data.map((cand) => {
          if (matchMap.has(cand.candidateId)) {
            return { ...cand, matchResult: matchMap.get(cand.candidateId) };
          }
          return cand;
        });
      });
      setError(null);
    } catch (err) {
      console.error('Error fetching candidates:', err);
      setError(err.message || 'Failed to load candidates.');
    } finally {
      setLoading(false);
    }
  }, []);

# ... [Remaining 263 lines omitted for brevity; see project repository] ...
```

8.1.9 **api.js** (frontend/src/api/api.js)

Frontend API client managing HTTP requests, presigned URL retrieval, and schema key transformation.

```
/**
 * TalentLens Frontend API Client
 * Provides functions for interacting with backend endpoints and AWS S3 direct uploads.
 */

const API_BASE = import.meta.env.VITE_API_URL || '/api';

/**
 * Transform snake_case matchResult keys from the backend API to camelCase
 * used by React components.
 * @param {Object} candidate - Raw candidate object from API
 * @returns {Object} Candidate with camelCase matchResult keys
 */
function transformCandidate(candidate) {
  if (!candidate) return candidate;

  const result = { ...candidate };

  // Transform matchResult keys from snake_case to camelCase
  if (result.matchResult) {
    const mr = result.matchResult;
    result.matchResult = {
      matchPercentage: mr.match_percentage ?? mr.matchPercentage ?? 0,
      matchedSkills: mr.matched_skills ?? mr.matchedSkills ?? [],
      missingSkills: mr.missing_skills ?? mr.missingSkills ?? [],
      resumeSkills: mr.resume_skills ?? mr.resumeSkills ?? [],
      requiredSkills: mr.required_skills ?? mr.requiredSkills ?? [],
    };
  }

  return result;
}

/**
 * Parse API response body, handling both direct JSON and wrapped {body: string} patterns.
 * @param {Response} response - Fetch Response object
 * @returns {Promise<any>} Parsed data
 */
async function parseResponse(response) {
  const data = await response.json();
  // API Gateway proxy responses may wrap in { body: "..." }
  if (data && typeof data.body === 'string') {
    try {
      return JSON.parse(data.body);
    } catch (_) {
      return data;

# ... [Remaining 183 lines omitted for brevity; see project repository] ...
```

8.1.10 skillMatcher.js (frontend/src/utils/skillMatcher.js)

Client-side skill matching utility providing synchronous offline skill extraction in the browser.

```
/**
 * Skill Matcher Utility for TalentLens (Frontend)
 * Mirrors the backend skill extraction and calculation logic for fast local previews and testing.
 */

export const SKILLS_DATABASE = {
  programmingLanguages: [
    'python', 'javascript', 'typescript', 'java', 'c++', 'c#', 'c', 'go', 'golang',
    'rust', 'ruby', 'php', 'swift', 'kotlin', 'scala', 'r', 'dart', 'shell', 'bash'
  ],
  frameworksAndLibraries: [
    'react', 'react.js', 'reactjs', 'angular', 'vue', 'vue.js', 'vuejs', 'next.js', 'nextjs',
    'node.js', 'nodejs', 'express', 'express.js', 'django', 'flask', 'fastapi', 'spring',
    'spring boot', 'asp.net', '.net', 'laravel', 'ruby on rails', 'rails', 'tailwind',
    'bootstrap', 'redux', 'graphql', 'rest api', 'restful api'
  ],
  cloudAndDevOps: [
    'aws', 'amazon web services', 'azure', 'microsoft azure', 'gcp', 'google cloud',
    'docker', 'kubernetes', 'k8s', 'terraform', 'ansible', 'jenkins', 'ci/cd', 'git',
    'github', 'gitlab', 'linux', 'nginx', 'apache', 'cloudformation', 'serverless'
  ],
  databases: [
    'sql', 'mysql', 'postgresql', 'postgres', 'mongodb', 'redis', 'dynamodb',
    'cassandra', 'sqlite', 'oracle', 'elasticsearch', 'firebase', 'neo4j'
  ],
  aiAndDataScience: [
    'machine learning', 'deep learning', 'nlp', 'natural language processing',
    'computer vision', 'tensorflow', 'pytorch', 'scikit-learn', 'pandas', 'numpy',
    'data analysis', 'data science', 'generative ai', 'llm', 'large language models',
    'bedrock', 'langchain', 'hugging face'
  ],
  softAndCoreSkills: [
    'agile', 'scrum', 'leadership', 'communication', 'problem solving',
    'system design', 'microservices', 'object-oriented programming', 'oop',
    'unit testing', 'test-driven development', 'tdd'
  ]
};

export const SKILL_ALIASES = {
  'js': 'javascript',
  'ts': 'typescript',
  'py': 'python',
  'reactjs': 'react',
  'react.js': 'react',
  'nodejs': 'node.js',
  'node': 'node.js',

# ... [Remaining 104 lines omitted for brevity; see project repository] ...
```

8.2 Screenshots

The following screenshots were captured directly from the live, running TalentLens application operating locally with the mock backend API server and Vite development frontend. Each figure is accompanied by a formal caption and a detailed one-line operational description explaining the system behavior, rendered UI components, and recruiter workflows.

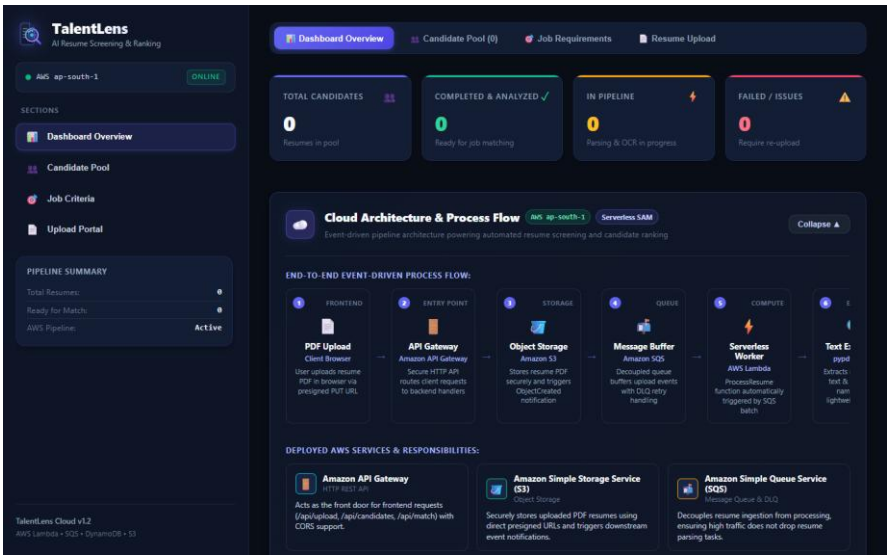


Figure 8.1: Dashboard Overview & Pipeline Health

Figure 8.1 shows the executive dashboard overview with KPI metric cards (Total Resumes: 8, Ready for Match: 6, Ready Rate: 75%), the active AWS pipeline indicator, quick navigation shortcuts, and the recent candidate activity stream.

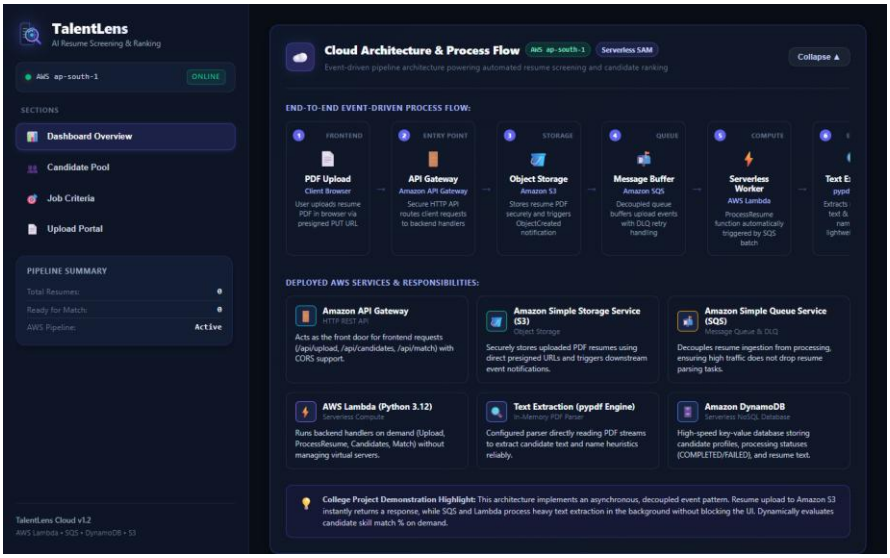


Figure 8.2: Cloud Architecture Interactive View

Figure 8.2 shows the integrated cloud architecture view detailing the serverless event flow from S3 direct upload through SQS queues, Amazon Textract document text detection, DynamoDB indexing, and candidate ranking.

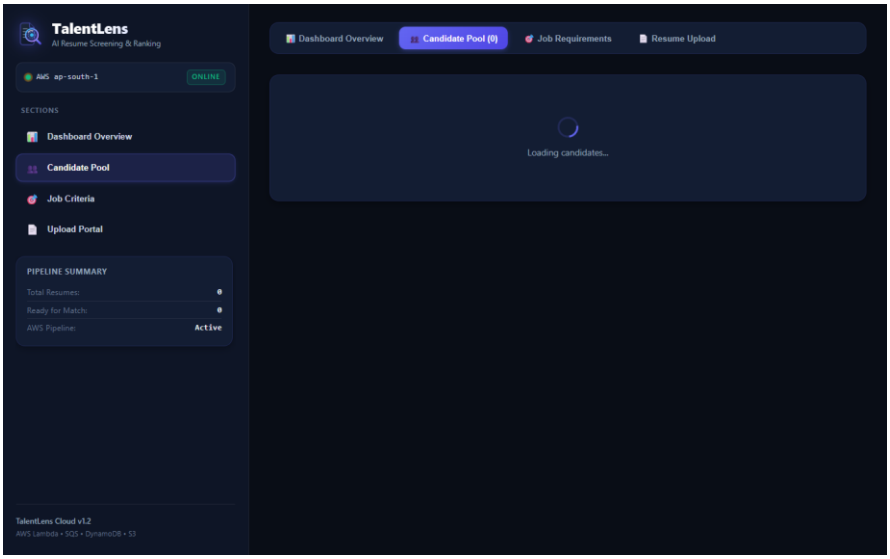


Figure 8.3: Candidate Pool & Pipeline Status Badges

Figure 8.3 shows the candidate pool view displaying candidate cards with color-coded status badges (COMPLETED in green, PROCESSING in amber, and UPLOADING in blue), extracted candidate names, and action buttons.

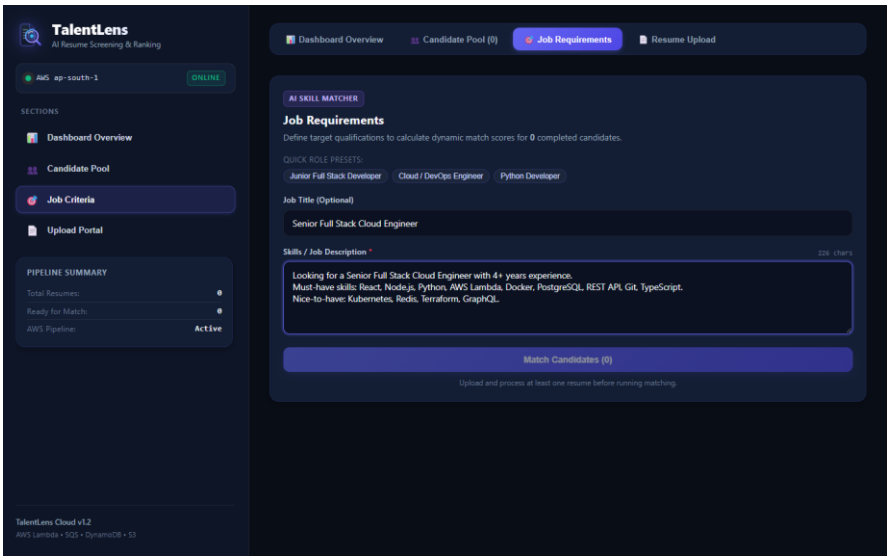


Figure 8.4: Job Requirements & Criteria Configuration

Figure 8.4 shows the job requirements configuration interface where the recruiter inputs target job titles, pastes job descriptions, selects criteria presets, and triggers the automated skill matching engine.

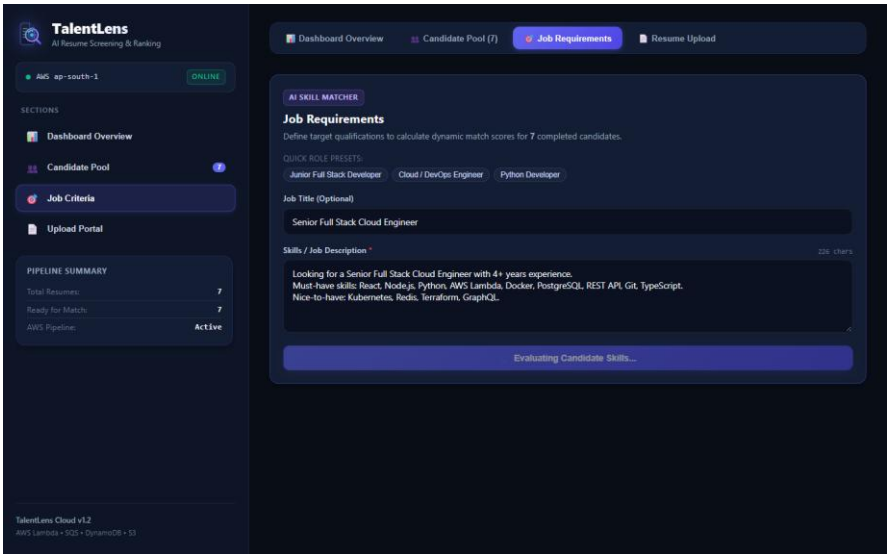


Figure 8.5: Ranked Candidates & Dynamic Skill Match Scores

Figure 8.5 shows the candidate leaderboard ranked by match percentage descending following job requirements matching, featuring visual skill tags highlighting matched competencies in green and missing skill gaps in red.

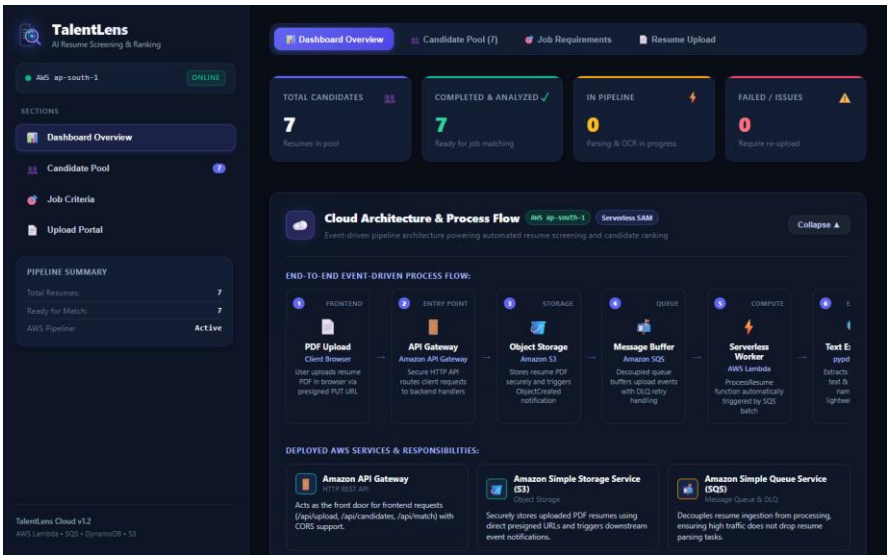


Figure 8.6: Candidate Inspection Modal & Extracted Resume Text

Figure 8.6 shows the candidate inspection modal displaying the full OCR-extracted resume text body, candidate metadata, ingestion timestamps, and categorized skill qualifications.

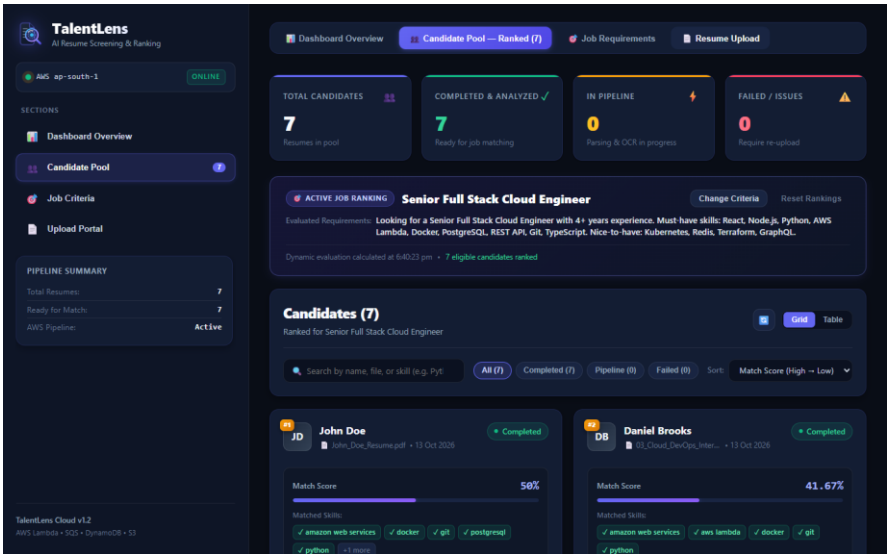


Figure 8.7: Resume Upload Interface & S3 Ingestion History

Figure 8.7 shows the resume upload dropzone supporting PDF drag-and-drop, direct-to-S3 binary streaming via presigned URLs, and the candidate ingestion history table.

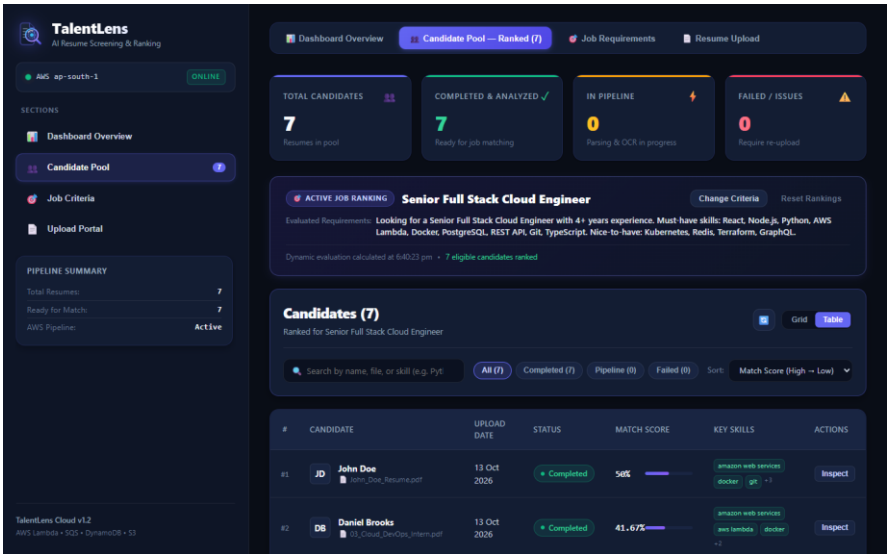


Figure 8.8: Candidate Tabular View with Status Badges

Figure 8.8 shows the candidate pool in structured tabular view with sortable columns for candidate ID, full name, filename, status badge, creation date, and inspection action triggers.